

Vignesh Venkatesh

vv2455@nyu.edu | github.com/Vignesh-Venkatesh | linkedin.com/in/vignesh-2k3 | vigneshvenkatesh.com

EDUCATION

New York University

New York, NY

Master of Science in Computer Science

Sep 2025 – May 2027

- **Relevant Coursework:** Parallel Programming, Advanced Database Systems, Real-Time Big Data Analytics

The Pennsylvania State University

State College, PA

Bachelor of Science in Computer Science

Aug 2021 – May 2025

- **Relevant Coursework:** Operating Systems, Computer Architecture, Advanced Data Structures and Algorithms, Theory of Computation, Discrete Mathematics, Calculus and Vector Analysis

WORK AND LEADERSHIP EXPERIENCE

Software Engineering Intern

Jun 2026 – Aug 2026

MiraPhoto

New York City, NY

- Built a **model feed feature** using **Expo, React Native, and TypeScript**, implementing image upload workflows and responsive feed rendering.
- Designed and implemented the **backend architecture** supporting **model uploads, image storage, retrieval, and display**.
- Reduced **image loading latency** by **approximately 85%** through **optimized image delivery** and application data flows.
- Investigated and methodically resolved **high-priority P0 production issues**, improving application stability and release quality.
- Bootstrapped a **CI/CD pipeline** using **YAML workflows and GitHub Actions**, automating builds and deployments through a **self-hosted Mac Mini runner**.

Distributed Control Systems Intern

May 2024 – Jun 2024

Yokogawa Electric

Abu Dhabi, U.A.E.

- Learned **SEBOL**, Yokogawa's proprietary **C-based control language**, and began developing **production code within one week**.
- Developed **alarm suppression logic**, a **step function**, and a **moving average function block** for industrial plant monitoring systems.
- Supported project engineering workflows using **GitHub Issues, GitHub Actions, and project planning tools** to prioritize bugs and coordinate work across a **3–5 member engineering team**.

Digital Design Course Learning Assistant

Aug 2023 – May 2025

The Pennsylvania State University

State College, PA

- Moderated course discussions, provided technical feedback to students, **graded 200+ assignments weekly**, and collaborated with faculty and teaching assistants to develop **digital design course materials**.

PROJECTS

DNS Resolver — Blog | GitHub

- Implemented an **iterative DNS resolver in Go** from scratch following **RFC 1035**, parsing binary DNS packets across a multi-hop resolution chain.
- Handled **CNAME traversal, nameserver fallback** when glue records were unavailable, **compression pointer resolution**, and circular pointer detection.
- Wrote **dependency-free unit tests** for the query builder, serializer, and parser, identifying **offset and packet truncation errors** before integration.
- **Technical Skills:** Go, DNS, Network Protocols, Binary Serialization, Unit Testing

Dynamic Memory Allocator — GitHub

- Built a **malloc/free-style dynamic memory allocator in C++**, implementing custom heap allocation, deallocation, splitting, and block coalescing.
- Developed a **segregated free-list structure** that organized memory blocks by size class, improving allocation efficiency and reducing fragmentation.
- Enforced **16-byte alignment** for allocated and free blocks while maintaining block metadata and heap consistency.
- **Technical Skills:** C++, Memory Management, Data Structures, Systems Programming

MovieBoxd — Demo | GitHub

- Built a **full-stack movie discovery and tracking platform** supporting authentication, ratings, reviews, and user watchlists.
- Designed **RESTful backend services** using **Hono and PostgreSQL** to manage user, review, and movie-related data.
- Implemented authentication with **Supabase Auth** and integrated the **TMDB API** to retrieve real-time movie metadata, ratings, and cast information.
- **Technical Skills:** React, TypeScript, Hono, PostgreSQL, Supabase, REST APIs, TMDB API

TECHNICAL SKILLS

- **Languages:** Go, Python, TypeScript, JavaScript, C, C++, SQL, HTML, CSS
- **Backend:** Node.js, Hono, Express.js, REST APIs, Better Auth
- **Frontend:** React, React Native, Expo, Tailwind CSS, DaisyUI
- **Databases:** PostgreSQL, SQLite, Supabase, MongoDB
- **Developer Tools:** Git, GitHub Actions, Linux, Docker